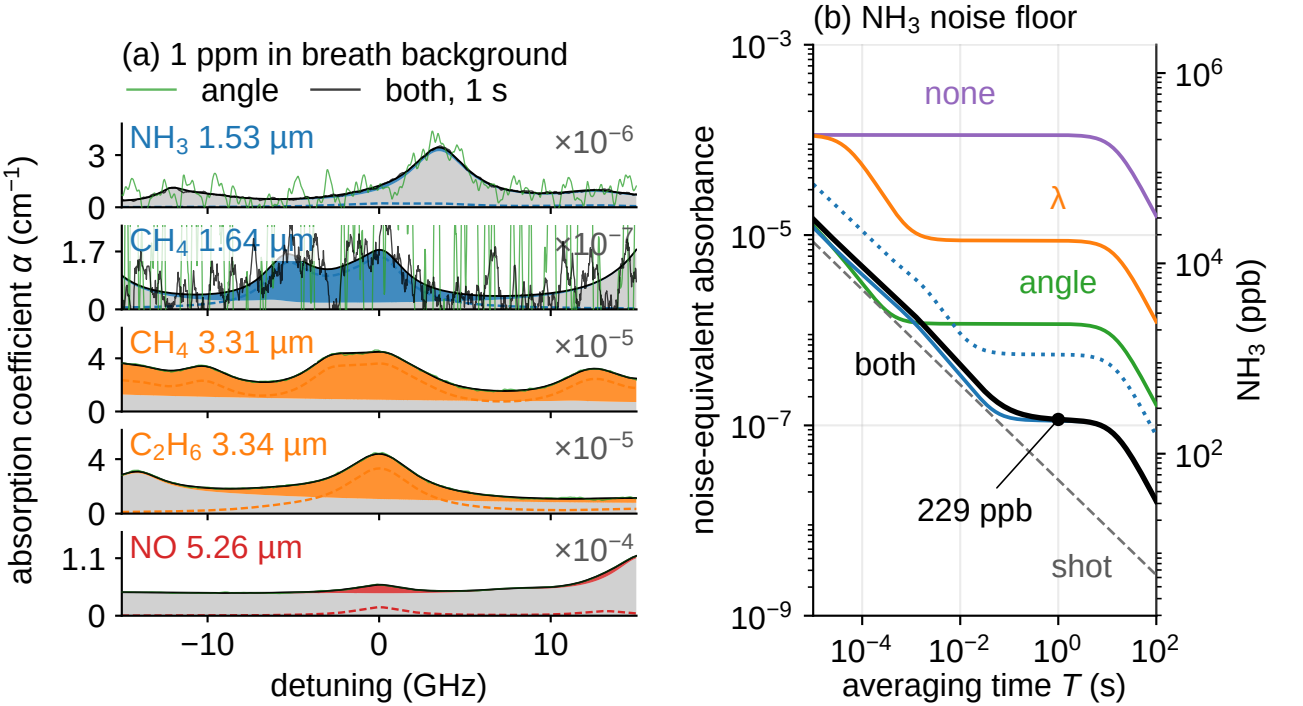

(a) 1 ppm in breath background
angle
both, 1 s
NH3 1.53 μm
×10−6
CH4 1.64 μm
×10−7
CH4 3.31 μm
×10−5
C2H6 3.34 μm
×10−5
NO 5.26 μm
×10−4
absorption coefficient α (cm−1)
detuning (GHz)
(b) NH3 noise floor
none
λ
angle
both
229 ppb
shot
noise-equivalent absorbance
NH3 (ppb)
averaging time T (s)